

DR. GITANJALI E. GNANADESIKAN

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Updated: April 15, 2026

EDUCATION

University of Arizona

August 19, 2023

PhD in Biological Anthropology

Minor in Cognitive Science

Dissertation: *Genetic and Endocrine Contributions to Dog Behavior and Cognition in a Working Dog Population*

Committee: Evan L. MacLean, Stacey R. Tecot, Bridgett M. vonHoldt, Mary A. Peterson

GPA: 4.0

University of Arizona

May 10, 2019

MA in Anthropology

Report: *Estimating the Heritability of Cognitive Traits Across Dog Breeds Reveals a Heritable Cooperative-Communicative Factor*

Adviser: Evan L. MacLean

GPA: 4.0

Princeton University

June 3, 2014

AB in Ecology and Evolutionary Biology, *magna cum laude*.

Thesis: *Exploring the Canine Methylome: The Impacts of Domestication on the Regulatory Genome*

Adviser: Bridgett M. vonHoldt

GPA: 3.72

ACADEMIC APPOINTMENTS

Emory University

August 2023 – Present

Postdoctoral Research Fellow with Marcela Benítez, Department of Anthropology

Atlanta, GA

- Emory FIRST (Fellowship in Research and Science Teaching)/NIH IRACDA Fellow*
- Project: Form and function of oxytocin in capuchins
- Affiliate at the Language Research Center, Georgia State University, conducting capuchin experiments
- Affiliate and instructor in the Biology Department at Spelman College (2024-2025)

FELLOWSHIPS

2023 – 2025*	Emory Fellowship in Research and Science Teaching/NIH IRACDA Fellow (\$238,000)
2023	University of Michigan Society of Fellows – Declined (\$207,000)
2018 – 2023	NSF Graduate Research Fellowship (\$138,000)
2017 – 2018	Andrew Carnegie University Fellow – University of Arizona first-year fellowship (\$63,311)
2014 – 2015	Teaching Fellow – Teach for China

*Originally awarded through 2027. The NIH IRACDA program was terminated on April 2, 2025.

† Co-first authors, * Undergraduate mentee, ** Graduate mentee, ^ Postbac mentee

21. Bray, Emily E., Laura E.L.C. Douglas, Kerinne M. Levy, **Gitanjali E. Gnanadesikan**, Daniel J. Horschler, Brenda S. Kennedy & Evan L. MacLean (in press). Maternal style and early rearing environment influence puppy behaviour and cognition. *Animal Behaviour*. Preprint: <https://www.biorxiv.org/content/10.64898/2025.11.29.691282v2>
20. **Gnanadesikan, Gitanjali E.**, Emily E. Bray, Kerinne M. Levy, Laura E.L.C. Douglas, Daniel J. Horschler, Stephanie Hargrave, Brenda S. Kennedy, Marina M. Watowich, Noah Snyder-Mackler, & Evan L. MacLean (2026). Characterizing the Heritability of Cognitive and Behavioral Traits across Development in Domestic Dogs. *Royal Society Open Science*, 13 (2): 241918. doi: 10.1098/rsos.241918
19. **Gnanadesikan, Gitanjali E.†**, Katherine M. King†, Elizabeth Carranza, Abigail Flyer^, Gianna Ossello^, Paige Smith^, Netzin Steklis, H. Dieter Steklis, C. Sue Carter, Jessica Connelly, Melissa Barnett, Nancy Gee, Stacey R. Tecot & Evan MacLean (2024). Effects of Human-Animal Interaction on Salivary and Urinary Oxytocin Concentrations in Children and Dogs. *Psychoneuroendocrinology*, 169 (Nov 2024), 107147. [Editor's Choice] doi: 10.1016/j.psyneuen.2024.107147
18. **Gnanadesikan, Gitanjali E.**, Emily E. Bray, Erica N. Cook^, Kerinne M. Levy, Laura E.L.C. Douglas, Brenda S. Kennedy, Stacey R. Tecot & Evan L. MacLean (2024). Basal Plasma Oxytocin & Fecal Cortisol Concentrations are Highly Heritable and Associated with Individual Differences in Behavior & Cognition in Dog Puppies. *Hormones & Behavior*, 165 (Sep 2024), 105612. [Featured on Journal Cover] doi: 10.1016/j.yhbeh.2024.105612
17. **Gnanadesikan, Gitanjali E.**, Elizabeth Carranza, Katherine M. King, Abigail Flyer^, Gianna Ossello^, Paige Smith^, Netzin Steklis, H. Dieter Steklis, Jessica Connelly, Melissa Barnett, Nancy Gee, Stacey R. Tecot & Evan MacLean (2024). Glucocorticoid response to naturalistic interactions between children and dogs. *Hormones & Behavior*. 161 (May 2024), 105523. doi: 10.1016/j.yhbeh.2024.105523
16. MacLean, Evan L., Elizabeth Carranza, **Gitanjali E. Gnanadesikan**, Katherine M. King, Alicia Allen, Linnea Linde-Krieger, Ruth Feldman, Rosemary C. White-Traut, Elizabeth A. D. Hammock, C. Sue Carter, Gareth Leng, Stacey R. Tecot, & Aleeca F. Bell (2024). Neuropeptide Y is an Analytically Robust Surrogate Biomarker for Oxytocin. *Psychoneuroendocrinology*, 161 (March 1, 2024): 106951. [Editor's Choice] doi: 10.1016/j.psyneuen.2023.106951
15. **Gnanadesikan, Gitanjali E.**, Dhriti Tandon, Emily E. Bray, Brenda S. Kennedy, Stavi Tennanbaum, Evan L. MacLean & Bridgett M. vonHoldt (2023). Transposons in the Williams-Beuren Syndrome critical region are associated with social behavior in assistance dogs. *Behavior Genetics*. doi: 10.1007/s10519-023-10166-7
14. ManyDogs Project, Julia Espinosa, Jeffrey R. Stevens, Daniela Alberghina, Harley E. E. Alway, Jessica D. Barela, Michael Bogese, Emily E. Bray, Daphna Buchsbaum, Sarah-Elizabeth Byosiore, Molly Byrne, Camila M. Cavalli, Leah M. Chaudoir*, Courtney Collins-Pisano, Hunter J. DeBoer, Laura E. L. C. Douglas, Shany Dror, Marina V. Dzik, Beverly Ferguson, Laura Fisher, Hannah C. Fitzpatrick, Marianne S. Freeman, Shayla N. Frinton, Maeve K. Glover, **Gitanjali E. Gnanadesikan**, Joshua E. P. Goacher, Marta Golańska, C.-N. Alexandrina Guran, Elizabeth Hare, Brian Hare, Mia Hickey*, Daniel J. Horschler, Ludwig Huber, Hoi-Lam Jim, Angie M. Johnston, Julianne Kaminski, Debbie M. Kelly, Valerie A. Kuhlmeier, Lily Lassiter, Lucia Lazarowski, Jennifer Leighton-Birch, Evan L. MacLean, Kamila Maliszewska, Vito Marra, Lane I. Montgomery, Madison S. Murray, Emma K. Nelson, Ljerka Ostojić, Shennai G. Palermo, Anya E. Parks Russell, Madeline H. Pelgrim, Sarita D. Pellowe, Anna Reinholz, Laura A. Rial, Emily M. Richards, Miriam A. Ross, Liza G. Rothkoff, Hannah Salomons, Joelle K. Sanger, Laurie Santos, Angelina R. Schirle, Shania J. Shearer, Zachary A.

- Silver, Jessica M. Silverman, Andrea Sommesse, Tiziana Srdoc, Hannah St. John-Mosse, Angelica C. Vega, Kata Vékony, Christoph J. Völter, Carolyn J. Walsh, Yasmin A. Worth, Lena M. I. Zipperling, Bianka Żołędzewska, Sarah G. Zylberfuden (2023). ManyDogs 1: A multi-lab replication study of dogs' pointing comprehension. *Animal Behavior & Cognition*, 10(3): 232-286. Preprinted registered report: 10.31234/osf.io/f86jq. Final doi: 10.26451/abc.10.03.03.2023
13. ManyDogs Project, Daniela Alberghina, Emily E. Bray, Daphna Buchsbaum, Sarah-Elizabeth Byosiére, Julia Espinosa, **Gitanjali E. Gnanadesikan**, C.-N. Alexandrina Guran, Elizabeth Hare, Daniel J. Horschler, Ludwig Huber, Valerie A. Kuhlmeier, Evan L. MacLean, Madeline H. Pelgrim, Bryan Perez, Dana Ravid-Schurr, Liza Rothkoff, Courtney L. Sexton, Zachary A. Silver & Jeffrey R. Stevens (2023). ManyDogs Project: A Big Team Science Approach to Investigating Canine Behavior and Cognition. *Comparative Cognition & Behavior Reviews*, 18, 59–77. doi: 10.3819/CCBR.2023.180004
12. Salomons, Hannah, Kyle Smith, Megan Callahan-Beckel, Margaret Callahan, Kerinne Levy, Brenda S. Kennedy, Emily Bray, **Gitanjali E. Gnanadesikan**, Daniel J. Horschler, Margaret Gruen, Jingzhi Tan, Philip White, Bridgett M. vonHoldt, Evan MacLean & Brian Hare (2023). Response to Hansen Wheat et al.: Additional analysis further supports the early emergence of cooperative communication in dogs compared to wolves raised with more human exposure. *Learning & Behavior*. doi: 10.3758/s13420-023-00576-2
11. **Gnanadesikan, Gitanjali E.**, Elizabeth Hammock, Stacey R. Tecot, Rebecca J. Lewis, Russ Hart, C. Sue Carter & Evan L. MacLean (2022). What are oxytocin assays measuring? Epitope mapping, metabolites, and comparisons of wildtype & knockout mouse urine. *Psychoneuroendocrinology*, 143, 105827. doi: 10.1101/2022.03.03.482682
10. Horschler, Daniel J., Emily E. Bray, **Gitanjali E. Gnanadesikan**, Molly Byrne, & Evan L. MacLean (2022). Dogs re-engage human partners when joint social play is interrupted. *Animal Behaviour*, 183, 159–168. doi: 10.1016/j.anbehav.2021.11.007.
9. **Gnanadesikan, Gitanjali E.**, Elizabeth A. D. Hammock, Stacey R. Tecot, C. Sue Carter & Evan L. MacLean (2021). Specificity of Plasma Oxytocin Immunoassays: A Comparison of Commercial Assays and Sample Preparation Techniques Using Oxytocin Knockout and Wildtype Mice. *Psychoneuroendocrinology*, 132, 105368. doi: 10.1016/j.psyneuen.2021.105368
8. Salomons, Hannah, Kyle Smith, Megan Callahan-Beckel, Margaret Callahan, Kerinne Levy, Brenda S. Kennedy, Emily Bray, **Gitanjali E. Gnanadesikan**, Daniel J. Horschler, Margaret Gruen, Jingzhi Tan, Philip White, Bridgett M. vonHoldt, Evan MacLean & Brian Hare (2021). Cooperative communication with humans evolved to emerge early in dogs. *Current Biology*, 31(14), 3137-3144.e11. doi: 10.1101/2021.01.12.425620
7. Bray, Emily E., **Gitanjali E. Gnanadesikan**, Daniel J. Horschler, Kerinne M. Levy, Brenda S. Kennedy, Tom R. Famula & Evan L. MacLean (2021). Early-emerging and highly-heritable sensitivity to human communication in dogs. *Current Biology*, 31(14), 3132–3136.e5. doi: 10.1016/j.cub.2021.04.055
6. Bray, Emily E., Margaret E. Gruen, **Gitanjali E. Gnanadesikan**, Daniel J. Horschler, Kerinne M. Levy, Brenda S. Kennedy, Brian A. Hare & Evan L. MacLean (2021). Dog cognitive development: A longitudinal study across the first two years of life. *Animal Cognition*, 24(2), 311–328. doi: 10.1007/s10071-020-01443-7.
5. **Gnanadesikan, Gitanjali E.**, Brian Hare, Noah Snyder-Mackler, Josep Call, Julianne Kaminski, Ádám Miklósi & Evan MacLean (2020). Breed differences in dog cognition associated with brain-expressed genes and neurological functions. *Integrative and Comparative Biology*, 60(4), 976–990. doi: 10.1093/icb/icaa112

4. Bray, Emily E., Margaret E. Gruen, **Gitanjali E. Gnanadesikan**, Daniel J. Horschler, Kerinne M. Levy, Brenda S. Kennedy, Brian A. Hare & Evan L. MacLean (2020). Cognitive characteristics of 8-to-10-week-old assistance dog puppies. *Animal Behaviour*, 166, 193-206. doi: 10.1016/j.anbehav.2020.05.019
3. **Gnanadesikan, Gitanjali E.**, Brian Hare, Noah Snyder-Mackler & Evan L. MacLean (2020). Estimating the Heritability of Cognitive Traits Across Dog Breeds Reveals Highly Heritable Inhibitory Control and Communication Factors. *Animal Cognition*, 23(5), 953–964. doi: 10.1007/s10071-020-01400-4
2. **Gnanadesikan, Gitanjali E.**, William D. Pearse & Allison K. Shaw (2017). Evolution of mammalian migrations for refuge, breeding, and food. *Ecology and Evolution*, 7(15), 5891–5900. doi: 10.1002/ece3.3120
1. Janowitz Koch, Ilana L., Michelle M. Creek, Michael J. Thompson, Kerry A. Deere-Machemer, Jun Wang, Lionel Duarte, **Gitanjali E. Gnanadesikan**, Eskender L. McCoy, Liudmilla Rubbi, Daniel R. Stahler, Matteo Pellegrini, Elaine A. Ostrander, Robert K. Wayne, Janet S. Sinsheimer & Bridgett M. vonHoldt (2016). The concerted impact of domestication and transposon insertions on methylation patterns between dogs and grey wolves. *Molecular Ecology*, 25(8), 1838–1855. doi: 10.1111/mec.13480

MANUSCRIPTS IN PREPARATION OR REVIEW

† Co-first authors, * Undergraduate mentee, ** Graduate mentee, ^Postbac mentee

King, Katherine M., Allison Hays, Arielle X. Liu, **Gitanjali E. Gnanadesikan**, Rebecca J. Lewis, Stacey R. Tecot (in revision). Exploratory study of oxytocin and male affiliation in Verreaux's sifaka (*Propithecus verreauxi*) in Kirindy Mitea National Park, Madagascar.

Gnanadesikan, Gitanjali E., Sarah A. Kovalaskas**, Sierra M. V. Simmons**, Sarah F. Brosnan, Marcela Benítez (in prep). Measuring Pro-8 Oxytocin in Tufted Capuchins (*Sapajus apella*): Challenges and Considerations.

Mistry, Arianna†^, Elizabeth Whiteside†*, **Gitanjali E. Gnanadesikan**, Sarah Brosnan, Marcela Benítez (in prep). Uncovering Menopause in Tufted Capuchin Monkeys (*Sapajus apella*): Analyzing the Relationship between Estradiol, Aging, and Behavioral Estrus in a Captive Population.

Gnanadesikan, Gitanjali E., Emily E. Bray, Noah Snyder-Mackler, Daniel J. Horschler, Stephanie Hargrave, Laura E.L.C. Douglas, Brenda S. Kennedy, Marina M. Watowich & Evan L. MacLean (in prep). *Genetic architecture of cognitive and behavioral traits in dogs reveals deep polygenicity and conserved links to human psychological traits.*

OTHER PUBLICATIONS

Gnanadesikan, Gitanjali, Toneisha Stubbs, Valerie Haftel, Triscia Hendrickson & Tiffany Oliver. Feds use DEI excuse to kill scientist training program in Atlanta. The Atlanta Journal-Constitution. Oct 17, 2025. <https://www.ajc.com/opinion/2025/10/feds-use-dei-excuse-to-kill-scientist-training-program-in-atlanta/>

Gnanadesikan, Gitanjali E., Emily E. Bray, Kerinne M. Levy, Daniel J. Horschler, Margaret E. Gruen, Brenda S. Kennedy, Brian Hare, and Evan L. MacLean. (2023) Dog Cognitive Development Battery Methods. *OSF*. October 20. doi:10.17605/OSF.IO/CSK3A.

Gnanadesikan, Gitanjali E. (2014) Exploring the Canine Methyome: The Impacts of Domestication on the Regulatory Genome. Thesis (Senior)-Princeton University, 2015. <http://arks.princeton.edu/ark:/88435/dsp011544bp26d>

Gnanadesikan, Gitanjali E. (2012) Identifying PRDM9 binding sites in meiotic recombination hotspots. *Summer and Academic Year Student Reports*. Paper 2371. <http://mouseion.jax.org/strp/2371>

HONORS & AWARDS

- 2026 Emory University Postdoctoral Association Travel Award (\$450)
- 2023 School of Anthropology Graduate Teaching Award (\$500)
- 2022 University of Arizona Herbert E. Carter Travel Award for interdisciplinary work (\$600)
- 2022 University of Arizona Graduate College Travel Grant (\$500)
- 2022 School of Anthropology Graduate Student Travel Award (\$800)
- 2021 University of Arizona Graduate College Travel Grant (\$500)
- 2020 PEO Scholar Award (\$15,000)
- 2019 University of Arizona Graduate College Travel Grant (\$500)
- 2018 University Fellows Professional Development Award (\$500)
- 2018 School of Anthropology Summer Award: Traditions, Transitions & Treasures Fund (\$2,000)
- 2018 Graduate and Professional Student Council Research and Project Grant (\$1,000)
- 2018 Graduate and Professional Student Council Travel Grant (\$750)
- 2017 NSF GRFP Honorable Mention
- 2014 Elected to the Sigma Xi scientific honors society
- 2014 Porter '52 EEB Research Fund conference award to present thesis work (\$750)
- 2013 Princeton EEB award to attend the International Canine and Feline Genomics conference (\$900)
- 2013 John T. Bonner Senior Thesis Fund grant for summer thesis work (\$4,400)

INVITED TALKS – ACADEMIC

- Biology Department at Spelman College. *The Biological Bases of Behavior in Canines and Primates*. February 19, 2026.
- Psychology Department at Hunter College. *The Biological Bases of Behavior & Cognition in Canines and Primates*. February 11, 2026.
- Biology Department at St. Olaf College. *The Biological Bases & Evolution of Canine Behavior & Cognition*. December 4, 2025.
- Psychology Department Colloquium at University of California Berkeley. *The Biological Bases & Evolution of Canine Behavior & Cognition*. November 10, 2025.
- Biology Department at Kalamazoo College. *The Biological Bases & Evolution of Canine Behavior & Cognition*. November 6, 2025.
- Biology Department at West Chester University. *The Biological Bases & Evolution of Canine Behavior*. October 31, 2025.

- Anthropology Department at Saint Mary's College of Maryland. Visiting Scholar, Early Career Series. *What Makes an Individual? How Genetics, Hormones, and the Environment Shape Your Dog's Behavior*. October 27-29, 2025.
- Neuroscience Institute at The University of Chicago. *Oxytocin, Social Behavior & Monkey Business*. July 15, 2025.
- Social & Cognitive Origins Group at Johns Hopkins University. *Oxytocin & Cognition in Canines & Primates: Prosocial, Anxiolytic, or None of the Above?* May 20, 2025.
- Canine Cognition Center at Boston College. *Genetic and Endocrine Contributions to Dog Behavior and Cognition in a Working Dog Population*. March 19, 2024.
- Social Cognition and Primate Behavior Lab at Emory University. *How do Genetics & Hormones Contribute to Behavior & Cognition and their Evolution?* March 27, 2023.
- Department of Psychology at Reed College. *What Makes an Individual? Using Genetics & Hormones to Explore Dog Behavior and Cognition*. November 21, 2022.
- Behavioral Neuroscience and Comparative Psychology seminar series at Arizona State University's Psychology Department. *Exploring the Genetic Bases of Behavior and Cognition in Dogs*. September 21, 2022.
- Comparative Cognition Lab at Yale University. *Exploring the Genetic Underpinnings of Dog Cognition*. February 19, 2021.

CONFERENCE PRESENTATIONS

22. **Gnanadesikan, Gitanjali E.**, Nicole M. Furgala, Arianna Mistry, Sarah F. Brosnan & Marcela E. Benítez. *Endogenous Oxytocin Manipulation in Tufted Capuchins (Sapajus apella) Decreases Intergroup Aggression but Does Not Affect Cooperation in a Group Paradigm*. Oral presentation at the 33rd Annual International Comparative Cognition Conference. Montreal, Canada. April 15–18, 2026.
21. **Gnanadesikan, Gitanjali E.**, Nicole M. Furgala, Arianna Mistry, Sarah F. Brosnan & Marcela E. Benítez. *Endogenous Oxytocin Manipulation in Captive Tufted Capuchins (Sapajus apella) Alters Social Behavior*. Oral presentation at the Animal Behavior Society 2025 Meeting. Baltimore, MD. July 8–12, 2025.
20. **Gnanadesikan, Gitanjali E.**, Emily E. Bray, Kerinne M. Levy, Laura E.L.C. Douglas, Daniel J. Horschler, Stephanie Hargrave, Brenda S. Kennedy, Marina M. Watowich, Noah Snyder-Mackler, & Evan L. MacLean. *Characterizing the Heritability of Cognitive and Behavioral Traits across Development in Domestic Dogs*. Oral presentation in the Warder Clyde Allee Symposium at the Animal Behavior Society 2024 Meeting. London, Ontario, Canada. June 26–29, 2024.
19. **Gnanadesikan, Gitanjali E.**, Emily E. Bray, Kerinne M. Levy, Laura E.L.C. Douglas, Daniel J. Horschler, Stephanie Hargrave, Brenda S. Kennedy, Marina M. Watowich, Noah Snyder-Mackler, & Evan L. MacLean. *Characterizing the Heritability of Cognitive and Behavioral Traits across Development in Domestic Dogs (Canis lupus familiaris)*. Oral presentation at the 31st Annual International Comparative Cognition Conference. Albuquerque, NM. April 10–13, 2024.
18. **Gnanadesikan, Gitanjali E.**, Emily E. Bray, Kerinne M. Levy, Laura E.L.C. Douglas, Daniel J. Horschler, Stephanie Hargrave, Brenda S. Kennedy & Evan L. MacLean. *Characterizing the Heritability of Cognitive and Behavioral Traits Across Development in a Population of Assistance Dogs*.

Oral presentation at the South Eastern Evolution and Human Behavior 2023 Meeting. Atlanta, GA. October 27, 2023.

17. **Gnanadesikan, Gitanjali E.**, Emily E. Bray, Erica N. Cook, Stacey R. Tecot & Evan L. MacLean. *Heritability and Behavioral Associations with Oxytocin and Cortisol Concentrations in Dog Puppies*. Oral presentation at the Animal Behavior Society 2023 Meeting. Portland, OR. July 12–15, 2023.
16. **Gnanadesikan, Gitanjali E.**, Dhriti Tandon, Emily E. Bray, Evan L. MacLean, & Bridgett M. vonHoldt. *Do dogs have elements of Williams-Beuren Syndrome? Transposons, behavior & training success in assistance dogs*. Oral presentation at the 30th Annual International Comparative Cognition Conference (Finalist in Ron Weisman Outstanding Student Presentation Competition). Melbourne, FL. April 12–15, 2023.
15. **Gnanadesikan, Gitanjali E.**, Dhriti Tandon, Emily E. Bray, Evan L. MacLean & Bridgett M. vonHoldt. *Williams-Beuren Syndrome, Behavior & Cognition, and Training Success in Assistance Dogs*. Presented at the Animal Behavior Twitter Conference 2023. January 18, 2023. https://twitter.com/g_gnanadesikan/status/1615718189664731136
14. **Gnanadesikan, Gitanjali E.**, Dhriti Tandon, Emily E. Bray, Evan L. MacLean & Bridgett M. vonHoldt. *Do dogs have elements of Williams-Beuren Syndrome? Transposons, behavior & training success in a population of assistance dogs*. Oral presentation at the Canine Science Conference 2022 Meeting. Hamilton, NY. October 7–9, 2022.
13. **Gnanadesikan, Gitanjali E.**, Dhriti Tandon, Emily E. Bray, Evan L. MacLean & Bridgett M. vonHoldt. *Do dogs have Williams-Beuren Syndrome? Transposons, behavior & training success in a population of assistance dogs*. Oral presentation at the Animal Behavior Society 2022 Meeting. San José, Costa Rica. July 20–23, 2022.
12. **Gnanadesikan, Gitanjali E.**, Elizabeth A. D. Hammock, Stacey R. Tecot, Rebecca J. Lewis & Evan L. MacLean. *Developing a new extraction method to minimize interference in immunoassay of urinary oxytocin*. Recorded talk for the International Society of Wildlife Endocrinology Virtual Conference 2021. August 16–17, 2021.
11. **Gnanadesikan, Gitanjali E.**, Emily E. Bray, Evan L. MacLean & Bridgett M. vonHoldt. *Williams-Beuren Syndrome in Dogs? Genetic Variation, Social Behavior, and Assistance Dog Success*. Oral presentation at the East Coast Canine Cognition Workshop. New Haven, CT. April 23–24, 2021.
10. **Gnanadesikan, Gitanjali E.**, Julia Espinosa & ManyDogs. *ManyDogs 1: An International Collaborative Approach to Pointing Comprehension in Domestic Dogs*. Presented at the Animal Behaviour Twitter Conference. January 27, 2021. <https://twitter.com/ManyDogsProject/status/1354550089918767105>
9. **Gnanadesikan, Gitanjali E.** & Evan L. MacLean. *Breed Differences in Heritable Cognitive Traits Associated with Brain-Expressed Genes and Neurological Functions in Dogs*. Presented at the Animal Behaviour Twitter Conference. January 27, 2021. https://twitter.com/g_gnanadesikan/status/1354547287112880134
8. **Gnanadesikan, Gitanjali E.**, Brian Hare, Noah Snyder-Mackler & Evan L. MacLean. *Exploring the Genetic Bases of Breed Differences in Dog Cognition*. Oral presentation at the East Coast Workshop on Canine Cognition. New Haven, CT. February 15–16, 2020.
7. **Gnanadesikan, Gitanjali E.**, Brian Hare, Noah Snyder-Mackler, and Evan L. MacLean. *Estimating the Heritability of Cognitive Traits Across Dog Breeds*. Oral presentation at the University of Arizona–Arizona State University Cognitive Science Conclave. Tucson, AZ. December 7, 2019.

6. **Gnanadesikan, Gitanjali E.**, Brian Hare, Noah Snyder-Mackler, and Evan L. MacLean. *Estimating the Heritability of Cognitive Traits Across Dog Breeds Reveals Highly Heritable Inhibitory Control and Cooperative-Communicative Factors*. Oral presentation at the Southwestern Association of Biological Anthropologists Annual Meeting. Tempe, AZ. November 1–2, 2019.
5. **Gnanadesikan, Gitanjali E.**, Brian Hare, Noah Snyder-Mackler, Evan L. MacLean. *Estimating the Heritability of Cognitive Traits Across Dog Breeds Reveals a Heritable Cooperative-Communicative Factor*. Oral presentation at the International Canine Science Conference. Phoenix, AZ. October 18–20, 2019.
4. **Gnanadesikan, Gitanjali E.**, Brian Hare & Evan L. MacLean. *Estimating the Heritability of Cognitive Traits Across Dog Breeds*. Oral presentation at the 26th Annual International Comparative Cognition Conference. Melbourne, FL. April 10–14, 2019.
3. **Gnanadesikan, Gitanjali E.**, Daniel J. Horschler & Evan L. MacLean. *Social Cues and Hormonal Profiles Over Development in Wolf Puppies*. Poster presentation at the Graduate and Professional Student Council Research Showcase. Tucson, AZ. February 13, 2019.
2. **Gnanadesikan, Gitanjali E.** & Evan L. MacLean, *Estimating the Heritability of Cognitive Traits*. Oral presentation at the East Coast Workshop on Canine Cognition. New Haven, CT. November 10–11, 2018.
1. **Gnanadesikan, Gitanjali E.** & Bridgett M. vonHoldt. *Exploring the Canine Methylome: The Impact of Domestication on the Regulatory Genome*. Poster presentation at the annual meeting for the Society for Integrative and Comparative Biology. Austin, TX. January 3–7, 2014

INTRAMURAL TALKS

Emory FIRST Seminar Series. *Form and Function of Oxytocin in Capuchins*. October 8, 2024.

OTHER ABSTRACTS

Bray, Emily E., Laura E.L.C. Douglas, **Gitanjali E. Gnanadesikan**, Daniel J. Horschler, Brenda S. Kennedy, Kerinne M. Levy & Evan L. MacLean. *Mothers matter: How early life experiences influence canine cognition and behavior*. Oral presentation by Emily Bray at the 31st Annual International Comparative Cognition Conference. April 10–13, 2024.

Liu, Arielle., Allison Hays, Katherine King, **Gitanjali E. Gnanadesikan**, Rebecca J. Lewis, Stacey R. Tecot. *Urinary Oxytocin and Aggression in Wild Verreaux's Sifaka (*Propithecus verreauxi*) in Kirindy Mitea National Park, Madagascar*. Poster presentation by Arielle Liu at the annual meeting of the American Society of Primatologists. June 20–23, 2023.

Bray, Emily E., Laura Douglas, **Gitanjali E. Gnanadesikan**, Daniel J. Horschler, Kerinne M. Levy, Brenda S. Kennedy & Evan L. MacLean. *Effects of early rearing environment on working dog puppy behavior and outcome*. Oral presentation by Emily Bray at the Canine Science Conference 2022 Meeting. October 7–9, 2022.

King, Katherine, Allison Hay, Arielle X. Liu, **Gitanjali E. Gnanadesikan**, Rebecca J. Lewis, & Stacey Tecot. *Investigating urinary oxytocin and affiliative behavior in wild male *Propithecus verreauxi* at Kirindy Mitea National Park, Madagascar*. Poster presentation by Katherine King at the annual meeting of the American Society of Primatologists. August 25–28, 2022

King, Katherine, Allison Hay, Arielle X. Liu, **Gitanjali E. Gnanadesikan**, Rebecca J. Lewis, & Stacey Tecot. Oxytocin and affiliative behavior in male *P. verreauxi* at Kirindy Mitea National Park, Madagascar. Poster presentation by Katherine King at the Animal Behavior Society 2022 Meeting. July 20–23, 2022.

Bray, Emily E., **Gitanjali E. Gnanadesikan**, Daniel J. Horschler & Evan L. MacLean. *Heritable variation in dog social cognition*. Presented by Emily Bray at the Animal Behaviour Twitter Conference. January 27, 2021. <https://twitter.com/DrEmilyBray/status/1354554832749645826>

MacLean, Evan L., Emily E. Bray, **Gitanjali E. Gnanadesikan** & Daniel J. Horschler. *Associations between individual differences in cognition and training outcomes in assistance dogs*. Presented by Evan MacLean at the East Coast Workshop on Canine Cognition. February 15–16, 2020.

Bray, Emily E., **Gitanjali E. Gnanadesikan**, Daniel J. Horschler & Evan L. MacLean. *Early development and longitudinal stability of cognitive traits in working dogs*. Presented by Emily Bray at the East Coast Workshop on Canine Cognition. February 15–16, 2020.

MacLean, Evan L., **Gitanjali E. Gnanadesikan**, Emily E. Bray & Noah Snyder-Mackler. *Dog Diversity as a Natural Experiment in Cognitive Evolution*. Presented by Evan MacLean at the Society for Integrative and Comparative Biology Annual Meeting. January 3–7, 2020.

MacLean, Evan L., Emily E. Bray, **Gitanjali E. Gnanadesikan** & Daniel J. Horschler. *Ontogeny and heritability of cognitive and temperamental traits in an assistance dog population*. Presented by Evan MacLean at the International Canine Science Conference, October 18–20, 2019.

Bray, Emily E., **Gitanjali E. Gnanadesikan**, Daniel J. Horschler & Evan L. MacLean. *Early emerging cognition in 9-week-old puppies*. Presented by Emily Bray at the 26th International Conference on Comparative Cognition. April 10–14, 2019.

MacLean, Evan L., Emily E. Bray, **Gitanjali E. Gnanadesikan**, Daniel J. Horschler. *Heritability of cognitive traits in a pedigree dog population*. Presented by Evan MacLean at the 26th International Conference on Comparative Cognition. April 10–14, 2019.

UNIVERSITY TEACHING EXPERIENCE – INSTRUCTOR

BIO 365: Animal Behavior

Fall 2024

Co-instructor with Dr. India Nichols-Obande

Spelman College

- Topics: Sensory systems, communication, behavioral genetics, learning & cognition, cooperation, etc.
- Duties included: contributing to syllabus revisions, lecturing and designing in-class activities, writing exam questions, and grading.

NSCS 200: Fundamentals of Neuroscience and Cognitive Science

Spring 2023

Graduate Teaching Associate

University of Arizona

- Topics: Neuron structure and function, sensory systems and perception, motor systems, introduction to cognitive science and methods.

ANTH 265: Human Evolution

Fall 2022

Instructor of Record

University of Arizona

- Topics: Evolutionary theory, genetic & epigenetic inheritance, human diversity, cognitive evolution.

ANTH 170C2: Animal Minds*Graduate Teaching Associate*

Fall 2019

University of Arizona

- Topics: Sensory systems, associative processes, decision making, tool use, categorization, social cognition, social learning, etc.
- Led two weekly discussion sections; graded assignments and final papers for 59 students. Contributed to curriculum development, created grading rubrics, and designed activities (used in subsequent years).

UNIVERSITY TEACHING EXPERIENCE – GUEST LECTURES

BIOL 190: Genetics*Invited by Dr. Graham Read*

November 7, 2025; April 3, 2026

Macalester College

- Topic: genetic and breed differences in dog behavior and cognition.

ANTH 101: Introduction to Anthropology*Early Career Visiting Scholar, Invited by Dr. Caitlin McPherson*

October 28, 2025

St. Mary's College of Maryland

- Topic: Career paths, breed differences in dog behavior & cognition, domestication.

NEUR 404: Social Brain Networks*Invited by Dr. Aparna Shah*

September 24, 2024

Virginia Tech

- Topic: Genetic and endocrine approaches to studying social behavior and cognition.

ES 102: Introduction to Environmental Science*Invited by Dr. Chhaya Werner*

May 1, 2024; May 12, 2025

Southern Oregon University

- Topic: Animal behavior and domestication.

ANTH 326: Domestication*Invited by Dr. Martin Welker*

February 2, 2022

University of Arizona

- Topic: Breed differences and the genetic bases of dog behavior & cognition.

PSYC 3362: Mind of a Dog*Invited by Dr. Angie Johnston*

September 10, 2020; September 9, 2021

Boston College

- Topic: Breed differences and the genetic bases of dog cognition.

New Start Summer Program*Invited by Dr. Elizabeth Eklund*

August 2020; July 2021

University of Arizona

- Topic: Introduction to biological anthropology and comparative cognition for incoming undergraduates.

NSCS 320: Issues and Themes in Cognitive Science*Invited by Dr. Vicki Lai*

December 3, 2020

University of Arizona

- Topic: Comparative cognition, canine cognition, cognitive evolution.

UNDERGRADUATE THESIS ADVISING

2023 – 2024	Elizabeth Whiteside (Emory, Committe Member)	<i>Uncovering Menopause in Tufted Capuchin Monkeys (Sapajus apella): Analyzing the Relationship between Estradiol, Aging, and Behavioral Estrus in a Captive Population</i> *Winner of the Outstanding Senior award in Anthropology
2021 – 2023	Mia Hickey (UARizona, Grad Mentor)	<i>ManyDogs 1: Effect of Canine Engagement on Performance in a Point-Following Task</i> *Winner of the Nugent Award for undergraduate research in Psychology
2021 – 2023	Leah Chaudoir (UARizona, Grad Mentor)	<i>Do Dogs Truly Understand Pointing Cues? How Eye Contact Affects Choices</i>

ADDITIONAL TEACHING & MENTORING EXPERIENCE

2025 –	Peer mentoring circle organized by Animal Behavior Society’s WFAB
2023 –	Mentoring eight bioanthropology graduate students at Emory
2021 – 2025	Mentoring junior graduate students through the Women of Color in EEB program
2022 – 2024	Peer mentoring circles organized by Women of Color in EEB
2024	Undergrad mentor for Animal Behavior Society’s Charles H. Turner Award program
2024	Invited panelist for mentoring event at the Comparative Cognition Society Conference
2020 – 2023	Mentored a junior graduate student in the School of Anthropology
2020 – 2023	Mentored a first-generation/low-income Princeton undergraduate
2022 & 2023	Invited panelist for graduate school “Survival Skills and Ethics” class at UARizona
2021 – 2022	Peer mentoring circle with graduate students, org. by Animal Behavior Society
2020	Mentored an undergraduate research assistant in the ACCC
2019 – 2020	Peer editor for University of Arizona’s Fellowship Application Support Program
2018 – 2019	Mentored a first-year graduate student in the University Fellows Program
2017 – 2018	Mentored a high school student from the Arizona MESA program
2016 – 2017	Trained undergraduates for research activities at the ACCC
2016	Tutored and mentored elementary school students in Baltimore
2014 – 2015	Taught English and music to third and fifth graders in rural China
2011 – 2013	Student mentor for Princeton University’s Integrated Science Curriculum

OUTREACH – INVITED TALKS

Princeton Women’s Network & Princeton Club of Georgia. *How Do Our Pets Think? And How Do We Know?* November 13, 2025.

Stand Up For Science & Science for Georgia. *How Do Our Pets Think?* June 25, 2025.

Splendido Assisted Living. *How Do Our Pets Think? And How Do We Know?* February 8, 2022.

Arizona Senior Academy. *How Do Our Pets Think? And How Do We Know?* July 14, 2021.

Petminded. *Dognition & Genetics*. February 7, 2021.

OUTREACH – SCIENTIFIC EDUCATION

- 2026 – Skype a Scientist volunteer
- 2023 – Letters to a Pre-Scientist STEM penpal
- 2021 Featured scientist in National Geographic Kids book “Can’t Get Enough Dog Stuff” by Stephanie Gibeault & Moira Donohue
- 2019 – 2020 Science Olympiad: weekly volunteer at Mansfeld Middle School in Tucson
- 2017 – 2018 Arizona MESA: weekly volunteer with a science club at Flowing Wells High School
- 2017 Dog Days with the Dean: experimental demonstration for undergraduate students
- 2017 Office of Admissions: experimental demonstration for AP high school students

PAST RESEARCH EXPERIENCE

Laboratory for the Evolutionary Endocrinology of Primates (LEEP) 2017–2023
Research Assistant with Evan MacLean and Stacey Tecot Tucson, AZ

- Developed, validated, and performed extraction and immunoassay protocols for oxytocin, vasopressin, and cortisol in a variety of biological samples; trained others on these methods.
- Contributed to studies on: human-animal bond (NIH), affiliation in male sifaka (NSF), and effects of maternal style on puppy behavior and cognition (AKC).
- Species studied included canines, humans, mice, lemurs, and titi monkeys.

ManyDogs 2017–2023
Project Coordinator, Methods Team, Team Captain

- General project administration, infrastructure building, and team management.
- Led the methods and protocol development, integrating suggestions from participating labs, and developing consensus among diverse groups.
- Led data collection at the Arizona Canine Cognition Center: trained and mentored undergraduate experimenters.

Fieldwork with Captive Wolf Pups 2018
Graduate Student Wildlife Science Center, MN

- Socialized, conducted behavioral tests on, and collected biological samples from captive wolf puppies.

Arizona Canine Cognition Center (ACCC) 2016–2017
Laboratory Coordinator Tucson, AZ

- Collaborated with multiple institutions to develop a battery of cognitive tests to use on puppies.
- Conducted cognitive behavioral experiments with companion dogs in the Tucson area at the ACCC and puppies at Canine Companions for Independence.

Senior Thesis Research

2013–2014

*Student with Bridgett vonHoldt**Princeton, New Jersey*

- “Exploring the Canine Methylome: The Impacts of Domestication on the Regulatory Genome”
- Conducted computational analysis of genome-wide methylation data with the aim of discovering differences in genetic regulation between dogs and wolves.

The Jackson Laboratory

2012

*Summer Student with Kenneth Paigen**Bar Harbor, Maine*

- Worked in a molecular biology laboratory on a project to identify the binding sites of PRDM9, a zinc-finger protein involved in meiotic recombination.

Princeton Ecology and Evolutionary Biology Department

Summer 2011

*Princeton Environmental Institute Intern with Allison Shaw**Princeton, New Jersey*

- Compiled a database of mammalian migratory behavior and related factors and analyzed the database for patterns regarding motivations for migration, geographic distribution, and spatial patterns.

ACADEMIC SERVICE

2026	Session chair at the 33rd Annual International Comparative Cognition Conference
2025	Stand Up For Science volunteer, speaker & action hour leader
2015 –	Princeton alumni interviewer
2025 –	Animal Behavior Society Conference “Ally” (with active bystander training)
2024 – 2026	Southeastern Evolution and Human Behavior (SEEHB) steering committee
2024	Session chair at the 31st Annual International Comparative Cognition Conference
2024	Reviewer for the Animal Behavior Society’s Student Research Grants
2022	Session chair at the Canine Science Conference
2022	Volunteer judge for the Animal Behavior Society’s undergraduate poster contest
2020 – 2022	Organizer of the department’s bioanthropology journal club
2020 – 2021	Vice president for the Anthropology Graduate Students at the University of Arizona
2020	Member of the School of Anthropology’s anti-racism ad hoc committee
2019 – 2020	Co-organizer for a students of color community group in the School of Anthropology
2018 – 2020	Organizer of the Arizona Canine Cognition Center’s journal club
2020	Co-organizer of a student-led departmental discussion on racism and anti-racism
2020	Founder & organizer of weekly School of Anthropology “teatime”
2019 – 2020	Student representative on the department’s Curriculum & Scheduling Committee
2019 – 2020	Secretary for the Anthropology Graduate Students at the University of Arizona
2018 – 2019	Travel and research grant judge for the Graduate and Professional Student Council
2010 – 2011	Co-founder of and historian for the Women in Science Colloquium at Princeton

PROFESSIONAL DEVELOPMENT

- 2026 PI Crash Course: Skills for Future or New Lab Leaders by the Columbia SHARP Program
- 2025 Weaving the Future of Animal Behavior (WFAB) by the Animal Behavior Society
- 2024 “How to Teach” (full semester class for IRACDA fellows at Emory)
- 2024 Responsible Conduct of Research (NIH compliant)
- 2023 Leadership and Management in Action Program (LMAP)

AD HOC REVIEWER

Animal Cognition, Comprehensive Psychoneuroendocrinology, Ethology Ecology & Evolution, iScience, Journal of the Experimental Analysis of Behavior, PeerJ, Physiology & Behavior, Scientific Reports, Applied Animal Behaviour Science, Journal of Comparative Psychology, American Journal of Primatology, PLOS ONE, Evolutionary Anthropology, Domestic Animal Endocrinology, Neuroscience and Biobehavioral Reviews.

See my Web of Science profile for verified peer reviews: <https://www.webofscience.com/wos/author/record/I-3571-2016>

PROFESSIONAL MEMBERSHIPS

Current: Animal Behavior Society, Comparative Cognition Society, International Society of Wildlife Endocrinology, Sigma Xi Honor Society

Past: Cognitive Science Society (2021-2022), Association for Women in Science (2024-2025), American Association of Biological Anthropologists (2025-2026)

SELECTED MEDIA COVERAGE

“Dogs Release This ‘Love Hormone’ Around Their Favorite People.” *The Farmer’s Dog Digest*. <https://www.thefarmersdog.com/digest/dogs-release-this-love-hormone-around-their-favorite-people/>

“Man’s Best Friend.” *Utah Public Radio: Undisciplined*. <https://www.upr.org/post/undisciplined-mans-best-friend>

“Puppies are biologically wired to communicate with people.” *Radio New Zealand*. <https://www.rnz.co.nz/national/programmes/sunday/audio/2018800500/puppies-are-biologically-wired-to-communicate-with-people>

“Do Dog Breeds Differ in Cognitive Traits?” *Psychology Today Blog: Animal Minds*. <https://www.psychologytoday.com/us/blog/animal-minds/202008/do-dog-breeds-differ-in-cognitive-traits>

“What a Crowdsourced Study Taught Us About How Dogs Learn.” *Smithsonian Magazine*. <https://www.smithsonianmag.com/science-nature/how-much-dogs-intelligence-hereditary-180975448>

“What separates dogs and wolves? Researchers journey to Anoka County to find out.” *Minnesota Star Tribune*. <http://www.startribune.com/what-separates-dogs-and-wolves-researchers-journey-to-anoka-county-to-find-out/488199251/>

TECHNICAL STRENGTHS

Lab Skills:	Immunoassays, Solid-Phase Extraction, DNA Extraction, PCR
Computer Languages:	R, Bash
Also familiar with:	Python, C++, Arduino, MATLAB, HTML
Other tools:	High Performance Computing, LaTeX, RMarkdown, Git
Languages:	English (native), Mandarin (moderate fluency and literacy)